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A Thesis in the Partial Fulfillment of the Requirements

for the Award of Bachelor of Computer Science and Engineering (BCSE)

Department of Computer Science and Engineering
College of Engineering and Technology
IUBAT—International University of Business Agriculture and Technology

Spring 2025

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A Thesis in the Partial Fulfillment of the Requirements for the Award of Bachelor of Computer Science and Engineering (BCSE)

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Letter of Transmittal

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Abstract

Autonomous Vehicle (AV) perception systems and Intelligent Transportation Systems (ITS) are increasingly vital for urban traffic management, yet models trained on structured Western datasets frequently fail in developing regions like Bangladesh. As of mid-2026, the Dhaka Metropolitan Police (DMP) has begun integrating AI-driven surveillance to automate the detection of traffic violations, lane infractions, and illegal parking. However, the efficacy of these systems is fundamentally limited by the visual complexity of local traffic—which is dense, unstructured, and non-lane-based—featuring extreme occlusion and unpredictable, heterogeneous vehicle geometries, such as non-motorized rickshaws and custom CNG auto-rickshaws. To address this, we propose a modified YOLOv11/YOLO26m architecture integrated with attention modules specifically designed to handle complex backgrounds and dense traffic flow. Our model utilizes a Convolutional Block Attention Module (CBAM) and a custom Dynamic Channel Fusion Module (DCFM_v2) injected into the deep backbone to enhance feature discrimination. We curated a region-specific dataset of over 289,506 annotated instances—aligned with the operational requirements of the national Road Transport Act 2018—to ensure our system accounts for the unique transport profile of the region. Our empirical results demonstrate that while baseline models provide a foundation, our attention-guided architecture significantly improves recall in high-density, occluded scenes, providing a reliable technical roadmap for the nationwide expansion of AI-driven traffic enforcement. We conclude by discussing the role of this perception framework in supporting broader smart-city security and the digital prosecution of traffic violations in Bangladesh.

Acknowledgments

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Chapter 1. Introduction

1.1 Problem Background and Motivation

The integration of AI-based automated traffic enforcement in Dhaka as of mid-2026 marks a historic shift in Bangladesh's law enforcement and infrastructure monitoring. The Dhaka Metropolitan Police (DMP) is currently deploying AI-integrated surveillance to generate digital case files for violations such as red-light running, wrong-way navigation, and illegal parking. While this initiative signals a vital modernization of our Intelligent Transportation Systems (ITS), the underlying perception technology faces significant hurdles in generalizing to the unstructured realities of Bangladeshi roads.

Unlike the disciplined traffic streams found in structured Western datasets, Bangladeshi traffic is characterized by a heterogeneous mix of motorized and non-motorized transport—including rickshaws, CNG auto-rickshaws, easy-bikes, buses, and commercial trucks—all sharing the same spatial frame. The specific challenges for AI-based enforcement systems include:

- **Extreme Heterogeneous Traffic Mix:** The simultaneous presence of high-speed vehicles and non-motorized cargo units creates a classification requirement that standard global benchmarks fail to satisfy.
- **Severe Spatial Occlusion:** In dense Dhaka traffic, vehicles frequently overlap, making it difficult for standard object detectors to isolate individual targets for Automatic Number Plate Recognition (ANPR) or precise violation documentation.
- **Algorithmic Generalization:** As the government scales the installation of hundreds of AI-enabled cameras across Dhaka, perception models must be optimized to operate reliably despite visual clutter, varied monsoon lighting, and high local vehicle diversity

Our research addresses these challenges by developing a region-aware traffic perception pipeline. We argue that for AI-driven enforcement to succeed at the scale envisioned by national policy, the perception models must be engineered to handle the unique visual features of local road users and the high-occlusion environments prevalent in our cityscapes. By integrating attention-guided architectures, we provide a robust technical framework that directly supports the goals of modernizing the Road Transport Act 2018 enforcement through digital, evidence-based prosecution.

[Comment: Insert Figure 1.1: A high-contrast comparison image showing a structured Western traffic scene versus a dense, unstructured Dhaka traffic scene to highlight the 'occlusion challenge' and 'heterogeneous mix'.]

1.2 Research Objectives and Scope

Standard convolutional models default to detecting highly frequent, rigid shapes (like cars) and often suppress complex minority classes (like bicycles and pedestrians) due to non-maximum suppression (NMS) errors. To overcome this, our research scope covers:

- **Academic Validation:** Designing a CBAM-integrated YOLO architecture robust enough to be peer-reviewed and accepted for presentation at the ICEFT 2026 conference.
- **Class-Specific Excellence:** Improving detection accuracy on region-specific, structurally complex vehicles like Rickshaws and CNGs.
- **Dataset Engineering:** Curating a master dataset of 289,506 instances, implementing multi-threaded validation and remapping scripts, and enforcing video-wise splits to ensure true model generalization.
- **Deployment:** Engineering a live Streamlit web application to serve as a functional model demonstration portal for future ITS integration.

1.3 Thesis Organization

The remainder of this thesis is structured as follows: Chapter 2 provides a critical assessment of the literature regarding real-time object detection and attention mechanisms. Chapter 3 details our methodology, moving from raw video ingestion to our multi-threaded

QA pipeline. Chapter 4 explores our core architectural innovations, including the injection of CBAM and DCFM modules. Chapter 5 documents our academic milestone at ICEFT 2026. Chapters 6 and 7 chronicle our experimental marathons, from cloud-based baselines to high-resolution (1024px) local workstation runs. Chapter 8 analyzes the "SAHI-Pro" 4-head architecture. Chapter 9 presents our production software and demonstration tools. Finally, Chapter 10 concludes the work with a summary of our contributions and an outline of future research, specifically regarding NMS-free tracking and vector-based street risk assessment.

Chapter 2. Literature Review

2.1 Evaluation of Real-Time Object Detection

For the past few years, object detection relied heavily on two-stage networks, but the field underwent a paradigm shift when Redmon et al. introduced the original YOLO (You Only Look Once) architecture. The YOLO family evolved rapidly with new versions and improvements. Recently, Jocher et al. advanced YOLO further with YOLOv11, reducing parameter counts and improving deep feature extraction through novel C3k2 modules and C2PSA attention mechanisms

However, fundamental challenges remain in YOLO architectures, as they often struggle with unstructured data. Zhang et al. observe in comparative studies that baseline YOLOv11 models face limitations in detecting highly occluded or structurally complex small objects. When YOLOv11 is deployed in dense, overlapping clusters of vehicles—a common occurrence in the traffic monitoring initiatives currently being implemented by the Dhaka Metropolitan Police (DMP)—recall drops significantly because the baseline cannot separate highly occluded bounding boxes without architectural modification.

2.2 Attention Mechanisms in Computer Vision

When a neural network gets very deep, it often loses fine details of small or partially hidden objects. To fix this issue, an attention mechanism was introduced to guide the network "where" and "what" to focus on, such as the Squeeze-and-Excitation (SE) network by Hu et al., which effectively captured channel importance, but ignored spatial location information entirely. To overcome this limitation, Woo et al. introduced the (CBAM)

Convolutional Block Attention Module, which achieves better attention by applying channel attention first, then spatial attention sequentially

Recently, integration of CBAM into the YOLOv11 backbone has become a key research focus for handling complex environments. In July 2025, Garshasebi et al. integrated CBAM into YOLOv11 for remote sensing imagery, achieving a mean Average Precision (mAP@50) of 76.68% by focusing on critical regions and reducing background noise. Also, Khan et al. applied YOLOv11 with CBAM for pipeline defect detection and observed accuracy improvements compared to the baseline model. These studies prove CBAM is very effective at extracting weak geometric features from noisy backgrounds, highlighting important features even when objects are overlapping in dense scenes.

2.3 Perception in Unstructured Traffic: The South Asian Context

When it comes to Intelligent Transportation Systems (ITS) in developing regions, the research is still catching up. From a Bangladeshi perspective, traffic in cities like Dhaka and Gazipur is chaotic and unstructured, and standard Western datasets often fail to include native vehicles like rickshaws or CNGs. Early attempts at automated enforcement faced significant challenges in detecting unstructured traffic. For example, a 2021 study by Rahman et al. used an ensemble of YOLOv5 models on the Dhaka-AI dataset and reached an mAP of only 45.8%, which shows the extreme difficulty of detecting traffic in such conditions

Recently, the open-source community has provided improved resources, such as the Bangladeshi Traffic Flow Dataset by Islam et al.. In 2022, Rahman et al. fine-tuned deep convolutional networks to classify native Bangladeshi vehicles with improved accuracy.

Furthermore, Hossain et al. analyzed the conditions of rickshaw pullers in Dhaka city and highlighted the significant presence of non-motorized transport in urban traffic environments.

Earlier studies usually treat deep learning models as black boxes, using transfer learning without fixing the mathematical cause of bounding box suppression in dense traffic. Our research differs from those earlier studies by building on regional datasets while going beyond simple retraining. We actively modify the YOLOv11 architecture by injecting CBAM directly into the deep feature layers, which targets the extreme occlusion issues in South Asian traffic across a full, 9-class unstructured environment. This aligns with the national priority of the DMP to replace manual oversight with robust, AI-driven traffic enforcement technology capable of processing the unique visual challenges of Dhaka's roads.

Chapter 3. Research Methodology

This section details the research design and procedural framework utilized to develop our traffic perception system. Our methodology follows an end-to-end pipeline, moving from raw data ingestion and annotation standardization to the engineering of attention-guided deep learning architectures, and finally to system deployment and empirical validation. The framework is designed to address the specific visual and structural challenges of Bangladeshi traffic, ensuring the model remains fit for purpose in automated enforcement scenarios as envisioned by current national transportation infrastructure initiatives.

3.1 Recap of Research Questions

Our research is driven by the necessity to bridge the gap between structured traffic detection models and the chaotic, non-lane-based environment of Dhaka. The core research questions are:

- How can deep learning architectures be structurally modified to account for the extreme occlusion and scale variance of Bangladeshi traffic?
- Can the injection of spatial and channel attention mechanisms significantly improve the detection accuracy of structurally complex, minority-class vehicles (e.g., cycles and rickshaws) without sacrificing real-time inference speed?
- What are the computational limitations and architectural trade-offs when transitioning from cloud-based prototype training to high-resolution local workstation deployments?

3.2 Research Design and Data Pipeline

The heart of our methodology is an end-to-end research pipeline that ensures repeatability and rigor. This pipeline converts raw, heterogeneous traffic frames into a high-performance detection framework.

[Comment: Insert Figure 3.1: Flowchart showing the pipeline from Raw Frames -> VOC XML -> YOLO Conversion -> QA Scanning -> Video-Wise Splitting -> Ablation Analysis -> Deployment.]

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Figure 3.1 Figure Title.

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3.2.1 Dataset Curation and Preprocessing

We curated the **Dhaka_Traffic_Local Version 6** dataset, which consists of 28,159 valid images and 289,506 annotated instances. The annotation pipeline utilized custom Python scripts to parse XML/JSON labels, standardize coordinate formats to YOLO-normalized values, and conduct multi-threaded sanity checks to eliminate corrupted entries and enforce class parity across the 8-class taxonomy.

3.2.2 Experimental Rationale

Our decision to move from standard 640px cloud baselines to 1024px local workstation runs was rooted in the "small-object vanishing" problem. At standard resolutions, distant road users (like cycles) were represented by too few pixels for the convolutional kernels to extract meaningful structural features. By upscaling to 1024px, we forced the network to preserve these high-frequency details. Our rationale for rejecting the SAHI patch-training method was equally empirical: slicing full-frame traffic scenes into 640×640 patches destroyed the global spatial context, proving that for this specific dataset, full-frame high-resolution training is superior to sliced-inference approaches.

3.3 Evaluation of Methods and Limitations

No research method is perfect. Our high-resolution 1024px approach, while successful in recovering recall, required significant computational resources—resulting in training marathons lasting up to 33.82 hours per model. Additionally, while the custom YOLO26m-SAHI-Pro 4-head architecture provides a theoretical solution for small-object rescue, it increases the GFLOPs of the network. We accept these trade-offs as necessary to achieve the high recall required for automated enforcement and safety monitoring under the Road Transport Act 2018.

3.4 hardware and Experimental Configuration

Our experiments were conducted across diverse computational environments to ensure the proposed architectural modifications were both accurate and efficient for real-world edge deployment.

Table 2.2 Hardware and Platform Configuration for Experimental Phases.

Phase	Platform	GPU Class	Batch Size
Cloud Baseline	Kaggle	2x Tesla T4	32
Cloud Stability Run	Google Colab	1x Tesla T4	8
High-Res Marathon	Local Workstation	NVIDIA RTX 4000	8

This table summarizes the diverse computational environments used, ranging from dual-GPU Kaggle baselines and single-GPU Colab stability runs to intensive local RTX 4000 high-resolution marathons.

3.4.1 Cloud-Based Ablation

Initial iterations utilized Kaggle and Google Colab cloud platforms. The Colab environment was particularly significant for our research; due to frequent runtime disconnections, we engineered a custom "Resume Protocol" that dynamically re-injected our `DCFM_v2` modules into the Ultralytics backend, allowing us to flawlessly resume training from checkpoint tensors. This capability was instrumental in successfully completing the V5.2 master training runs without losing progress after cloud instance timeouts.

3.4.2 Local Workstation High-Resolution Marathons

To resolve the "small-object vanishing" issue observed at standard 640px resolutions, we pivoted to a local workstation equipped with an NVIDIA RTX 4000 (8GB VRAM). These marathons involved upscaling input images to 1024px, necessitating strict batch size adjustments and the use of SSD-based data transfers to prevent I/O bottlenecks.

Chapter 4. Results and Discussion

In this chapter, we present the empirical results of our custom-engineered YOLO architectures and critically discuss their performance within the context of Bangladeshi traffic perception. The results are organized to sequentially address our primary research questions: first, evaluating the impact of attention mechanisms on baseline models; second, demonstrating the effectiveness of high-resolution training on small-object recall; and third, analyzing the architectural trade-offs of the SAHI-Pro diagnostic framework.

4.1 Comparative Analysis of Model Architecture

To establish the efficacy of our proposed architectural modifications, we compared several iterations against the standard YOLOv11 baseline. Our results demonstrate that integrating spatial and channel attention (CBAM) significantly enhances detection precision in dense scenarios.

Table 4.1 Comparative Metrics for Proposed Architectural Iterations.

Model	Classes	Image Size	mAP@50
YOLOv11m (Baseline)	9	640px	75.10%
YOLOv11m + CBAM	9	640px	75.70%
YOLO26m	8	640px	48.20%

(Local Dhaka)			
YOLO26m (High-Res)	9	1024px	49.78%

As shown in Table 4.1, the YOLOv11m-CBAM model achieved a superior mAP of 75.70%, validating our approach for the ICEFT 2026 conference. However, when moving to the more complex Dhaka_Traffic_Local V6 dataset, the baseline mAP dropped, highlighting the extreme difficulty of the real-world operational environment.

4.2 Impact of High-Resolution Training

A key finding of our research is the positive correlation between input resolution and minority-class recall. By scaling our input to 1024px on a local NVIDIA RTX 4000 workstation, we observed a significant recovery in small-object detection, particularly for cycles and pedestrians.

While the global mAP reached 49.78%, the most significant measurable outcome was the increase in global box recall to 67.90%. This indicates that our architectural modifications successfully reduced the "small-object vanishing" effect, ensuring that even distant or partially occluded road users are identified—a requirement critical for the reliability of AI-driven traffic enforcement systems under the Road Transport Act 2018.

4.3 Discussion and Findings

Our results suggest that while attention mechanisms (CBAM/DCFM) are vital for filtering background clutter, they are not a substitute for spatial resolution in high-density environments. The failure of the SAHI patch-training experiment (mAP < 1%) served as a vital negative result, confirming that slicing traffic scenes destroys the long-range spatial context—such as the relationship between a bus and a nearby rickshaw—that is necessary for the model to make informed enforcement decisions.

Furthermore, the integration of our models into the broader Dhaka Metropolitan Police (DMP) automated enforcement initiative highlights the importance of real-time performance. Our attention-guided models maintain a latency of approximately 5.3 ms per frame, ensuring that the system can handle the high-speed video feeds required for digital prosecution without creating system bottlenecks.

4.3.1 Limitations and Future Directions

Despite the improvements in recall, precision plateaued due to severe class imbalance, specifically regarding rare transport variants. Future research should prioritize synthetic data generation for these minority classes to further refine the precision of automated violation documentation. Additionally, while our current model excels at detection, the integration of NMS-free tracking and vector-based Time-To-Collision (TTC) estimation remains a necessary next step for transitioning from passive monitoring to active, automated traffic safety enforcement

Chapter 5. Conclusion

This thesis has presented a robust, attention-guided deep learning framework tailored specifically to the unstructured and high-occlusion traffic environments of Dhaka, Bangladesh. By integrating Convolutional Block Attention Modules (CBAM) and custom Dynamic Channel Fusion Modules (DCFM_v2) into the YOLOv11 and YOLO26m architectures, we have successfully addressed critical limitations in baseline object detection models currently utilized in automated traffic surveillance initiatives.

5.1 Main Findings and Contributions

Our primary finding is that standard, pre-trained object detectors are insufficient for the heterogeneous and densely packed vehicle profiles prevalent in South Asian urban centers. Through our research, we achieved the following:

- **Architectural Innovation:** We demonstrated that the injection of attention mechanisms into deep feature layers allows for significant gains in detection precision, particularly for occluded or minority-class vehicles like rickshaws, which achieved a 90.5% mAP in our refined testing phase.
- **Dataset Robustness:** By curating the Dhaka_Traffic_Local Version 6 dataset—comprising 289,506 annotated instances—we established a benchmark that better represents local traffic density compared to global standards.
- **Systemic Solution:** Our ablation studies provided a definitive answer to the "small-object vanishing" problem, proving that high-resolution (1024px) training marathons outperform traditional patch-slicing methods (such as SAHI) which destroy global spatial context.

The significance of this work lies in its direct alignment with the enforcement goals of the Road Transport Act 2018. By providing a perception model capable of generating reliable digital evidence in chaotic environments, we offer a technical roadmap for the Dhaka Metropolitan Police (DMP) to scale their automated enforcement and digital prosecution capabilities effectively.

5.2 Main Findings and Contributions

While our research has significantly improved detection recall, several avenues remain for future exploration:

- **Precision Bottlenecks:** Despite our recall improvements, precision plateaued due to the structural similarity between various non-motorized transport variants. Future research should leverage synthetic data generation or GAN-based augmentation to further distinguish between these overlapping vehicle classes.
- **From Detection to Tracking:** Our current focus was on object detection. Future studies should integrate NMS-free tracking and vector-based Time-To-Collision (TTC) assessment to enable the system to detect and predict traffic violations in real-time, moving beyond static frame analysis.
- **Edge Deployment:** While our models achieve efficient inference speeds of ~5.3 ms/frame, further optimization through tensor quantization or pruning is recommended for deployment on lower-power edge devices at intersection sites.

in conclusion, this thesis proves that specialized architectural modifications are essential for the success of intelligent transportation systems in the Global South. By prioritizing spatial attention and high-resolution input, we have developed a resilient baseline that supports the national transition toward smarter, automated, and safer city infrastructure.

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